

A Contraction Criterion for the Gibbs Fixed-Point Iteration in Entropy-Regularized Time-Inconsistent MDPs

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Abstract

We give a rigorous convergence theorem for the value iteration

$$y_{n+1} = \Psi_\lambda(y_n)$$

associated with the Gibbs fixed-point map in the entropy-regularized, discrete-time, finite-state time-inconsistent Markov decision process of Bayraktar, Huang, Wang, and Zhou. The result is *partial progress*, not a full solution of the general convergence problem: it proves convergence only under additional first-moment summability, entropy-envelope, invariant-ball, and explicit contraction assumptions. Under these hypotheses the map is a strict contraction on an absorbing ball, the value iterates converge geometrically to a unique fixed point in that ball, and the associated Gibbs policies converge to a regular relaxed equilibrium. The criterion is quantitative and checkable; it is strongest when the controlled transition law has small action-diameter, delayed rewards have small action-oscillation, or the effective horizon is short.

1 Status of the result

Bayraktar, Huang, Wang, and Zhou [1] study relaxed equilibria for time-inconsistent Markov decision processes with general non-exponential discounting. In the entropy-regularized discrete-time model, equilibria are characterized by a Gibbs-form fixed-point operator. Their existence proof uses compactness and Brouwer's fixed-point theorem, rather than a convergence analysis of the Picard iteration.

The present note proves a convergence theorem for a natural value-space Picard iteration. It should be read with the following limitations in mind.

- (i) The result concerns the entropy-regularized discrete-time problem with a finite state space.
- (ii) It does not treat the unregularized set-valued fixed-point problem obtained as the entropy parameter tends to zero.
- (iii) It does not prove convergence under the full hypotheses of [1]; it imposes stronger first-moment and contraction assumptions.
- (iv) It gives a sufficient condition, not a necessary one. Failure of the displayed contraction inequality below does not imply divergence.

Thus the result is genuine partial progress on the fixed-point-iteration question, but it is not a complete solution of the authors' general problem.

2 Model and notation

Let

$$\mathcal{S} = \{1, \dots, d\}$$

be a finite state space. Let $U \subset \mathbb{R}^\ell$ be compact with

$$0 < m_U := \text{Leb}(U) < \infty.$$

For each action $u \in U$, let $p^u = (p_{ij}^u)_{i,j \in \mathcal{S}}$ be a stochastic matrix, and write $p_i^u \in \mathbb{R}^d$ for its i -th row. Let

$$f : \mathbb{N}_0 \times \mathcal{S} \times U \rightarrow \mathbb{R}$$

be the reward kernel, where the first variable denotes the delay from the current decision time. Let $(\delta_t)_{t \geq 0}$ be nonnegative discount weights, with $\delta_0 = 1$. We assume throughout that all maps in the action variable are Borel measurable.

For a probability density $\rho \in \mathcal{D}(U)$, where densities are taken with respect to Lebesgue measure, set

$$H(\rho) := - \int_U \rho(u) \log \rho(u) du$$

whenever the integral is well-defined in $(-\infty, \infty]$. Fix an entropy parameter $\lambda > 0$. For $y \in \mathbb{R}^d$, define the Gibbs density

$$\Gamma_\lambda(y, i)(u) := \frac{\exp(\lambda^{-1}(f(0, i, u) + p_i^u \cdot y))}{\int_U \exp(\lambda^{-1}(f(0, i, v) + p_i^v \cdot y)) dv}, \quad u \in U. \quad (1)$$

We write

$$\gamma_i^y := \Gamma_\lambda(y, i).$$

The transition matrix, delayed reward vector, and entropy vector induced by y are

$$P_y(i, j) := \int_U p_{ij}^u \gamma_i^y(u) du, \quad (2)$$

$$r_t^y(i) := \int_U f(t, i, u) \gamma_i^y(u) du, \quad (3)$$

$$h_y(i) := H(\gamma_i^y). \quad (4)$$

The value fixed-point map considered below is

$$\Psi_\lambda(y) := \sum_{k=0}^{\infty} P_y^k (r_{k+1}^y + \lambda \delta_{k+1} h_y), \quad (5)$$

whenever the series is absolutely convergent. The shift from k to $k+1$ reflects that Ψ_λ is the continuation value under the Gibbs policy chosen by the current value vector y .

We use the norms

$$\|x\|_\infty := \max_{i \in \mathcal{S}} |x_i|, \quad \|A\|_\infty := \max_{i \in \mathcal{S}} \sum_{j \in \mathcal{S}} |A_{ij}|.$$

Define the primitive bounds

$$F_t := \sup_{i \in \mathcal{S}, u \in U} |f(t, i, u)|, \quad t \geq 1,$$

$$B_t := \sup_{i \in \mathcal{S}} \operatorname{osc}_{u \in U} f(t, i, u), \quad t \geq 0,$$

$$\Delta := \sup_{i \in \mathcal{S}} \sup_{u, v \in U} \|p_i^u - p_i^v\|_1.$$

Here $0 \leq \Delta \leq 2$. The number Δ is the maximum action-diameter of the transition rows.

3 Assumptions

The convergence theorem uses the following assumptions.

A1. The current reward has finite action-oscillation:

$$B_0 < \infty.$$

A2. The delayed rewards and discount weights satisfy

$$S_F := \sum_{k=0}^{\infty} F_{k+1} < \infty, \quad S_B := \sum_{k=0}^{\infty} B_{k+1} < \infty,$$

$$T_F := \sum_{k=0}^{\infty} k F_{k+1} < \infty,$$

$$D_0 := \sum_{k=0}^{\infty} \delta_{k+1} < \infty, \quad D_1 := \sum_{k=0}^{\infty} k \delta_{k+1} < \infty.$$

The first moments T_F and D_1 are stronger than the summability needed merely to define values; they enter when comparing the powers P_y^k and P_z^k .

A3. There is a continuous nondecreasing function

$$E : [0, \infty) \rightarrow [0, \infty)$$

such that

$$\|h_y\|_{\infty} \leq E(\|y\|_{\infty}), \quad y \in \mathbb{R}^d. \quad (6)$$

Remark 3.1 (Entropy envelopes). The entropy envelope is separated as an assumption because the most useful form depends on the regularity of the action model. Under stronger geometric hypotheses, such as the uniform-cone type hypotheses used in the entropy estimates of [1], one can use sharper envelopes, often with logarithmic growth in $\|y\|_{\infty}$. The proof below only needs whatever valid envelope is available. Without such sharper estimates, a cruder linear envelope follows directly from bounded oscillation of the Gibbs potential: since

$$H(\gamma_i^y) = \log \int_U e^{(f(0, i, v) + p_i^v \cdot y)/\lambda} dv - \lambda^{-1} \int_U (f(0, i, u) + p_i^u \cdot y) \gamma_i^y(u) du,$$

one has

$$|H(\gamma_i^y)| \leq |\log m_U| + \lambda^{-1} (B_0 + \Delta \|y\|_{\infty}).$$

The logarithmic envelope gives a substantially stronger absorbing-ball criterion than this elementary linear estimate.

For $R > 0$, define

$$G(R) := S_F + \lambda D_0 E(R) \quad (7)$$

and

$$L_R := \frac{\Delta}{2\lambda} S_B + \frac{\Delta(B_0 + \Delta R)}{4\lambda} D_0 + \frac{\Delta^2}{2\lambda} (T_F + \lambda E(R) D_1). \quad (8)$$

4 Main theorem

Theorem 4.1 (Convergence on an invariant ball). Assume [A1.](#)–[A3.](#). Let $R > 0$ satisfy

$$G(R) \leq R \quad (9)$$

and

$$L_R < 1. \quad (10)$$

Then Ψ_λ is well-defined on

$$\mathcal{B}_R := \{y \in \mathbb{R}^d : \|y\|_\infty \leq R\},$$

maps $\mathcal{B}_R$ into itself, and is a strict contraction on $\mathcal{B}_R$ in the ℓ^∞ -norm. Consequently, there is a unique fixed point $y^* \in \mathcal{B}_R$, and for every $y_0 \in \mathcal{B}_R$, the iteration

$$y_{n+1} = \Psi_\lambda(y_n)$$

satisfies

$$\|y_n - y^*\|_\infty \leq L_R^n \|y_0 - y^*\|_\infty, \quad n \geq 0. \quad (11)$$

The Gibbs feedback policy

$$\pi^*(i) := \Gamma_\lambda(y^*, i), \quad i \in \mathcal{S},$$

is a regular relaxed equilibrium of the entropy-regularized MDP.

Theorem 4.2 (Global convergence from arbitrary value initialization). Assume the hypotheses of [Theorem 4.1](#). Suppose in addition that

$$G(r) < r, \quad r \geq R. \quad (12)$$

Then, for every $y_0 \in \mathbb{R}^d$, the iteration $y_{n+1} = \Psi_\lambda(y_n)$ enters $\mathcal{B}_R$ after finitely many steps and then converges geometrically to the fixed point y^* of [Theorem 4.1](#). Moreover, y^* is the unique fixed point of Ψ_λ in all of $\mathbb{R}^d$.

Corollary 4.3 (Convergence of the associated policy iteration). Let

$$\Phi_\lambda(\pi)(i) := \Gamma_\lambda(V_\lambda^\pi, i)$$

be the Gibbs policy-improvement map, where V_λ^π denotes the entropy-regularized continuation value under π . Suppose either the hypotheses of [Theorem 4.1](#) hold and $V_\lambda^{\pi_0} \in \mathcal{B}_R$, or the hypotheses of [Theorem 4.2](#) hold. Define

$$\pi_{n+1} = \Phi_\lambda(\pi_n), \quad n \geq 0,$$

and set $y_n := V_\lambda^{\pi_n}$. Then $y_n \rightarrow y^*$, and

$$\max_{i \in \mathcal{S}} \|\pi_{n+1}(i) - \pi^*(i)\|_{L^1(U)} \leq \frac{\Delta}{\lambda} \|y_n - y^*\|_\infty. \quad (13)$$

Thus the policy iterates converge in the statewise L^1 -distance to the regular relaxed equilibrium π^* .

5 Proofs

We first record three estimates for the Gibbs map.

Lemma 5.1 (Lipschitz estimate for Gibbs densities). *For all $i \in \mathcal{S}$ and $y, z \in \mathbb{R}^d$,*

$$\|\gamma_i^y - \gamma_i^z\|_{L^1(U)} \leq \frac{\Delta}{\lambda} \|y - z\|_\infty. \quad (14)$$

Consequently,

$$\|P_y - P_z\|_\infty \leq \frac{\Delta^2}{2\lambda} \|y - z\|_\infty \quad (15)$$

and, for every $t \geq 1$,

$$\|r_t^y - r_t^z\|_\infty \leq \frac{\Delta B_t}{2\lambda} \|y - z\|_\infty. \quad (16)$$

Proof. Fix $i \in \mathcal{S}$. For $s \in [0, 1]$, set

$$y_s := z + s(y - z), \quad a_s(u) := f(0, i, u) + p_i^u \cdot y_s,$$

and let

$$\gamma_s(u) := \frac{e^{a_s(u)/\lambda}}{\int_U e^{a_s(v)/\lambda} dv}.$$

Writing $b(u) := p_i^u \cdot (y - z)$, differentiation gives

$$\frac{d}{ds} \gamma_s(u) = \frac{1}{\lambda} \gamma_s(u) \left(b(u) - \int_U b(v) \gamma_s(v) dv \right).$$

Hence

$$\begin{aligned} \int_U \left| \frac{d}{ds} \gamma_s(u) \right| du &\leq \frac{1}{\lambda} \sup_{u, v \in U} |b(u) - b(v)| \\ &\leq \frac{1}{\lambda} \sup_{u, v \in U} \|p_i^u - p_i^v\|_1 \|y - z\|_\infty \\ &\leq \frac{\Delta}{\lambda} \|y - z\|_\infty. \end{aligned}$$

Integrating in s proves (14).

For the transition estimate, fix a row i and write

$$\mu(du) := (\gamma_i^y(u) - \gamma_i^z(u)) du.$$

The signed measure μ has total mass zero. Let $\mu = \mu^+ - \mu^-$ be its Jordan decomposition and set

$$m := \mu^+(U) = \mu^-(U) = \frac{1}{2} \|\gamma_i^y - \gamma_i^z\|_{L^1(U)}.$$

If $m = 0$, there is nothing to prove. Otherwise,

$$P_y(i, \cdot) - P_z(i, \cdot) = m \left(\int_U p_i^u \frac{\mu^+(du)}{m} - \int_U p_i^u \frac{\mu^-(du)}{m} \right).$$

Both integrals belong to the convex hull of $\{p_i^u : u \in U\}$, whose ℓ^1 -diameter is at most Δ . Thus

$$\|P_y(i, \cdot) - P_z(i, \cdot)\|_1 \leq \frac{\Delta}{2} \|\gamma_i^y - \gamma_i^z\|_{L^1(U)} \leq \frac{\Delta^2}{2\lambda} \|y - z\|_\infty.$$

Taking the maximum over i proves (15). The proof of (16) is the same scalar argument with the action-diameter of $u \mapsto f(t, i, u)$, namely B_t , in place of Δ . $\square$

Lemma 5.2 (Local Lipschitz estimate for the entropy). *If $\|y\|_\infty, \|z\|_\infty \leq R$, then*

$$\|h_y - h_z\|_\infty \leq \frac{\Delta(B_0 + \Delta R)}{4\lambda^2} \|y - z\|_\infty. \quad (17)$$

Proof. Fix $i \in \mathcal{S}$ and use the interpolation from the proof of Lemma 5.1. The entropy of γ_s can be written as

$$H(\gamma_s) = \log \int_U e^{a_s(v)/\lambda} dv - \lambda^{-1} \int_U a_s(u) \gamma_s(u) du.$$

Differentiating this identity and using $b(u) = p_i^u \cdot (y - z)$, we obtain

$$\frac{d}{ds} H(\gamma_s) = -\lambda^{-2} \text{Cov}_{\gamma_s}(a_s, b). \quad (18)$$

By Gruss' inequality,

$$|\text{Cov}(A, B)| \leq \frac{1}{4} \text{osc}(A) \text{osc}(B)$$

for bounded random variables A, B . Since $y_s \in \mathcal{B}_R$,

$$\text{osc}_{u \in U} a_s(u) \leq B_0 + \Delta R, \quad \text{osc}_{u \in U} b(u) \leq \Delta \|y - z\|_\infty.$$

Combining these bounds with (18) and integrating from $s = 0$ to $s = 1$ yields (17). $\square$

Lemma 5.3 (Contraction estimate for Ψ_λ). *For all $y, z \in \mathcal{B}_R$,*

$$\|\Psi_\lambda(y) - \Psi_\lambda(z)\|_\infty \leq L_R \|y - z\|_\infty. \quad (19)$$

Proof. For $k \geq 0$, set

$$a_k^y := r_{k+1}^y + \lambda \delta_{k+1} h_y.$$

By Lemmas 5.1 and 5.2, for $y, z \in \mathcal{B}_R$,

$$\|a_k^y - a_k^z\|_\infty \leq \left(\frac{\Delta B_{k+1}}{2\lambda} + \frac{\Delta(B_0 + \Delta R)}{4\lambda} \delta_{k+1} \right) \|y - z\|_\infty. \quad (20)$$

By the entropy envelope,

$$\|a_k^z\|_\infty \leq F_{k+1} + \lambda \delta_{k+1} E(R). \quad (21)$$

Since P_y and P_z are stochastic matrices, $\|P_y\|_\infty = \|P_z\|_\infty = 1$, and the telescoping identity gives

$$P_y^k - P_z^k = \sum_{m=0}^{k-1} P_y^m (P_y - P_z) P_z^{k-1-m}.$$

Therefore

$$\|P_y^k - P_z^k\|_\infty \leq k \|P_y - P_z\|_\infty \leq \frac{k \Delta^2}{2\lambda} \|y - z\|_\infty. \quad (22)$$

Using (5), (20), (21), and (22),

$$\begin{aligned} \|\Psi_\lambda(y) - \Psi_\lambda(z)\|_\infty &\leq \sum_{k=0}^{\infty} \|P_y^k (a_k^y - a_k^z)\|_\infty + \sum_{k=0}^{\infty} \|(P_y^k - P_z^k) a_k^z\|_\infty \\ &\leq \sum_{k=0}^{\infty} \|a_k^y - a_k^z\|_\infty + \sum_{k=0}^{\infty} k \|P_y - P_z\|_\infty \|a_k^z\|_\infty \\ &\leq \left[\frac{\Delta}{2\lambda} S_B + \frac{\Delta(B_0 + \Delta R)}{4\lambda} D_0 + \frac{\Delta^2}{2\lambda} (T_F + \lambda E(R) D_1) \right] \|y - z\|_\infty. \end{aligned}$$

The bracket is L_R . $\square$

Proof of Theorem 4.1. Let $y \in \mathcal{B}_R$. Since P_y is stochastic, $\|P_y^k\|_\infty = 1$. By Assumption A3.,

$$\sum_{k=0}^{\infty} \left\| P_y^k (r_{k+1}^y + \lambda \delta_{k+1} h_y) \right\|_\infty \leq \sum_{k=0}^{\infty} (F_{k+1} + \lambda \delta_{k+1} E(R)) = G(R) < \infty.$$

Thus $\Psi_\lambda(y)$ is well-defined and

$$\|\Psi_\lambda(y)\|_\infty \leq G(R) \leq R.$$

Hence $\Psi_\lambda(\mathcal{B}_R) \subseteq \mathcal{B}_R$. Lemma 5.3 and $L_R < 1$ show that Ψ_λ is a strict contraction on the complete metric space $(\mathcal{B}_R, \|\cdot\|_\infty)$. Banach's fixed-point theorem gives the unique fixed point $y^* \in \mathcal{B}_R$ and the rate estimate (11).

It remains to identify the fixed point with a regular relaxed equilibrium. Set $\pi^*(i) = \Gamma_\lambda(y^*, i)$. Since $y^* = \Psi_\lambda(y^*)$, the value of π^* is y^* :

$$V_\lambda^{\pi^*} = y^*.$$

For each state i , the density $\Gamma_\lambda(y^*, i)$ is the unique maximizer over $\rho \in \mathcal{D}(U)$ of

$$\int_U (f(0, i, u) + p_i^u \cdot y^*) \rho(u) du + \lambda H(\rho).$$

This is the usual variational characterization of the Gibbs density, obtained by strict concavity of $\rho \mapsto \int A\rho + \lambda H(\rho)$ and the normalization constraint $\int \rho = 1$. Replacing y^* by $V_\lambda^{\pi^*}$ gives exactly the one-step optimality condition for a regular relaxed equilibrium of the entropy-regularized problem. $\square$

Proof of Theorem 4.2. Let $r_n := \|y_n\|_\infty$. The estimate in the proof of Theorem 4.1, with R replaced by r_n , gives

$$r_{n+1} \leq G(r_n), \quad n \geq 0. \quad (23)$$

If $r_n \geq R$, then (12) implies $r_{n+1} < r_n$. Suppose, for contradiction, that the orbit never enters $\mathcal{B}_R$. Then (r_n) is decreasing and bounded below by R , so it converges to some $r_\infty \geq R$. Passing to the limit in (23), using the continuity of G , yields

$$r_\infty \leq G(r_\infty) < r_\infty,$$

a contradiction. Hence the orbit enters $\mathcal{B}_R$ after finitely many steps. Theorem 4.1 then gives geometric convergence from the entrance time onward.

If $\bar{y}$ is any fixed point with $\bar{r} := \|\bar{y}\|_\infty \geq R$, then $\bar{r} = \|\Psi_\lambda(\bar{y})\|_\infty \leq G(\bar{r}) < \bar{r}$, a contradiction. Thus no fixed point lies outside $\mathcal{B}_R$, and uniqueness inside $\mathcal{B}_R$ follows from Theorem 4.1. $\square$

Proof of Corollary 4.3. By definition of Φ_λ ,

$$\pi_{n+1}(i) = \Gamma_\lambda(y_n, i), \quad y_n = V_\lambda^{\pi_n}.$$

Therefore

$$y_{n+1} = V_\lambda^{\pi_{n+1}} = V_\lambda^{\Gamma_\lambda(y_n, \cdot)} = \Psi_\lambda(y_n).$$

The value convergence follows from Theorem 4.1 or Theorem 4.2. Finally, Lemma 5.1 gives

$$\|\pi_{n+1}(i) - \pi^*(i)\|_{L^1(U)} = \|\Gamma_\lambda(y_n, i) - \Gamma_\lambda(y^*, i)\|_{L^1(U)} \leq \frac{\Delta}{\lambda} \|y_n - y^*\|_\infty.$$

Taking the maximum over i proves (13). $\square$

6 A discounted-reward specialization

Suppose

$$f(t, i, u) = \delta_t g(i, u), \quad \delta_0 = 1,$$

where

$$G_g := \sup_{i,u} |g(i, u)| < \infty, \quad B_g := \sup_i \operatorname{osc}_{u \in U} g(i, u) < \infty,$$

and suppose $\sum_{t \geq 1} t \delta_t < \infty$. Then

$$S_F = G_g D_0, \quad S_B = B_g D_0, \quad T_F = G_g D_1, \quad B_0 = B_g.$$

The contraction constant becomes

$$L_R = \frac{\Delta B_g D_0}{2\lambda} + \frac{\Delta(B_g + \Delta R) D_0}{4\lambda} + \frac{\Delta^2}{2\lambda} (G_g D_1 + \lambda E(R) D_1). \quad (24)$$

Thus the iteration converges whenever an absorbing radius R satisfies $G(R) \leq R$ and the explicit quantity in (24) is below one; under $G(r) < r$ for all $r \geq R$, the convergence is global from arbitrary value initialization.

Remark 6.1 (Interpretation of the three terms). The first term in L_R controls the change in delayed reward caused by changing the Gibbs density. The second controls the entropy component of the regularized payoff. The third controls the change in the future state law through the matrix powers P_y^k . The first-moment sums T_F and D_1 appear only in this third term.

7 Conclusion

The criterion

$$G(R) \leq R, \quad L_R < 1$$

turns the Gibbs fixed-point map into a strict contraction on an invariant ball. With the absorbing strengthening $G(r) < r$ for $r \geq R$, the convergence is global. This provides a rigorous convergence theorem for the entropy-regularized value iteration and the associated Gibbs policy iteration in a quantitative small-action-dependence regime.

It does not settle the full convergence question in the generality of [1]. In particular, it leaves open convergence outside the contraction regime, the zero-entropy limit, and the continuous-time counterpart. The contribution is therefore best described as partial progress: it proves a clean sufficient condition under which the fixed-point iteration is mathematically justified.

References

- [1] E. Bayraktar, Y.-J. Huang, Z. Wang, and Z. Zhou. Relaxed equilibria for time-inconsistent Markov decision processes. *Mathematics of Operations Research*, 50(4):2666–2687, 2025. doi:[10.1287/moor.2023.0209](https://doi.org/10.1287/moor.2023.0209); arXiv:[2307.04227](https://arxiv.org/abs/2307.04227).